

RANDOM VARIABLE:

Discrete:

For discrete random variables we use Probability mass functions (PMF).

For a valid PMF we check following conditions

1) $P(X=a) = P_X(a)$

2) $0 \leq P_X(x) \leq 1$

3) $\sum_x P_X(x) = 1$

Example: 2 dice thrown.

$$S = \{ (1,1), (1,2), \dots, (6,6) \}$$

$$S_X = \{ 2, 3, 4, \dots, 12 \} \text{ sum of dots}$$

$$P_X(2) = 1/36, P_X(3) = 2/36, P_X(4) = 3/36,$$

$$P_X(5) = 4/36, P_X(6) = 5/36, P_X(7) = 6/36,$$

$$P_X(8) = 5/36, P_X(9) = 4/36, P_X(10) = 3/36,$$

$$P_X(11) = 2/36, P_X(12) = 1/36.$$

$$\sum P_X(x) = 1 \text{ (so valid PMF)}$$

PMF:

$$P_X(x) = \begin{cases} 1/36 & x = 2, 12 \\ 2/36 & x = 3, 11 \\ 3/36 & x = 4, 10 \\ 4/36 & x = 5, 9 \\ 5/36 & x = 6, 8 \\ \vdots & \vdots \\ 0 & \text{otherwise.} \end{cases}$$

Example 2:

$$S = \{H, T\}$$

$$\text{PMF} = \begin{cases} 1/2 & Y = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

$\Rightarrow$ Discrete random variable is defined by its range and PMF.

Expected value of R.V:

Using prev example 1:

$$\mu = E[X] = \sum_{\omega} x P_X(x)$$

$$(2)(1/36) + (3)(2/36) + (4)(3/36) + (5)(4/36) + (6)(5/36) \dots$$

$$\text{Ans} = 7.$$

Squared distance of RV from EV:

$$(X - E[X])^2$$

no constant, its random.

$$E[(X - E[X])^2] = \text{var}[X]$$

$$\text{var}[X] = E[X^2] - [E(X)]^2$$

$\Rightarrow$ variance can never be negative and 0.
if $\text{var} = 0$ then its not random.

Finding variance of previous ex. 1.

$$\Rightarrow E[X^2] = (2^2)(1/36) + (3^2)(2/36) + (4^2)(3/36) + (5^2)(4/36) + (6^2)(5/36) + (7^2)(6/36) \dots$$
$$= 54.83$$

$$54.83 - (7)^2 = 5.83 \text{ Ans}$$

$\Rightarrow$ CDF (Cumulative distribution function) is mostly used for continuous but we can also use it for discrete

$$F_X(a) = P_X(X \leq a)$$

$$\sum_{x=-\infty}^a P_X(x) \quad \sum_{x \leq a} P_X(x)$$

in prev example 1. prob less equal to 7 can be found by

$$P(X \leq 7) = \sum_{x=2}^7 P_X(x) \text{ or } 1 - P_X(8)$$

PROPERTIES FOR LINEAR OPERATION:

$$\mu = E[X]$$

$$\sigma^2 = \text{Var}[X]$$

$$E[aX] = a E[X]$$

$$E[X+b] = E[X] + b$$

$$E[aX+b] = aE[X] + b.$$

BERNOULLI RANDOM VARIABLE:

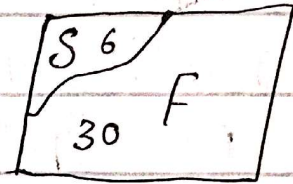
$$S = \{(1,1), \dots, (6,6)\}$$

RV = no. of successes in one.

$$\text{trial} = S_x = \{0, 1\}$$

Success = same no. of dots on both ~~sides~~ dice.

Failure = otherwise.



$$p_x(x) = \begin{cases} 30/36(1-p) & x=0 \\ 6/36(p) & x=1 \end{cases}$$

$$E(x) = p, \quad \text{Var}(x) = p(1-p)$$

DISCRETE UNIFORM R.V:

$$p_x(x) = 1/N$$

$$E(x) = \frac{1}{N} \sum_{i=1}^N a_i, \quad \text{Var}(x) = \frac{1}{N} \sum_{i=1}^N [a_i - E(x)]^2$$

$$S = \{H, T\}$$

$$S_x = \{0, 1\}$$

$$p_x(x) = \frac{1}{2}$$

R.V x shows number of success in N trials.

$$S_x = \{0, 1, 2, \dots, N\} \quad S_x = \{a_1, a_2, \dots, a_N\}$$

Binomial:

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$E(X) = np, \quad \text{var}(X) = np(1-p)$$

k successes in n trials.

$$S_X = \{0, 1, 2, 3, \dots, n\}$$

Geometric RV:

RV, X shows number of Bernoulli trials perform to get 1st success.

$$S_X = \{1, 2, 3, \dots, \infty\}$$

$$p_X(n) = p(1-p)^{n-1}$$

$$E(X) = \frac{1}{p}, \quad \text{var}(X) = \frac{1-p}{p^2}$$

Negative Binomial RV:

RV, X shows no. of Bernoulli trials perform to get r successes

$$S_X = \{r, r+1, r+2, \dots, \infty\}$$

$$p_X(n) = \binom{n-1}{r-1} p^r (1-p)^{n-r}$$

$$E(X) = \frac{r}{p}, \quad \text{var}(X) = r \left(\frac{1-p}{p^2} \right)$$

n trials to observe r successes.

Poisson RV: Type of discrete RV. No. of arrivals in given time frame.

$$S_x = \{0, 1, 2, \dots\}$$

$$p_x(k) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}$$

$$E(x) = \lambda t \quad \text{var}(x) = \lambda t$$

Continuous RV: Random variable x shows the temp

$$15 \leq x \leq 45$$

$$a \leq x \leq b, \quad x \in [a, b]$$

$$p_x(x=26) = 0$$

$$\rho = \frac{m}{V}$$

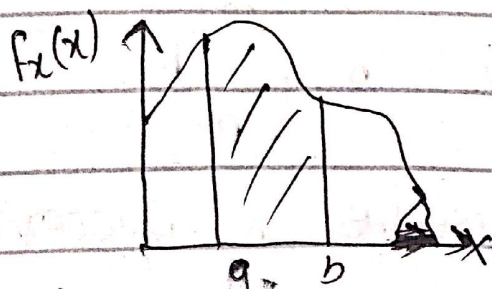
$$m = \int_x \int_y \int_z \rho(x, y, z) dx dy dz$$

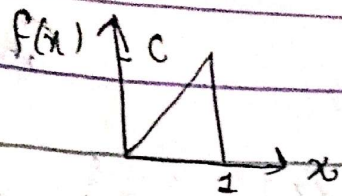
=> Prob Density Func.

$$i) f_x(x) \geq 0$$

$$ii) \int_{-\infty}^{\infty} f_x(x) dx = 1$$

$$iii) P_x(a \leq x \leq b) = \int_a^b f_x(x) dx$$





$c = ?$

$$1 = \frac{1}{2} \times c$$

$$\Delta = \frac{1}{2} b \times h$$

$$c = 2$$

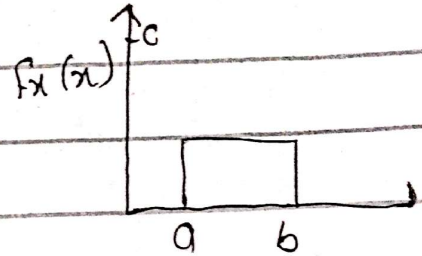
$$f_X(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

CONTINUOUS UNIFORM RV:

$$f_X(x) = \frac{1}{b-a} \quad a \leq x \leq b$$

$$E(X) = \frac{a+b}{2}$$

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$



$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$\text{Var}[X] = E[X^2] - (E[X])^2$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f_X(x) dx$$

EXPONENTIAL RV:

$$f_X(x) = \lambda e^{-\lambda x} \quad x \geq 0$$

$$E(X) = \frac{1}{\lambda}$$

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

GAUSSIAN RV:

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} \quad -\infty \leq x \leq \infty$$

$$E(X) = \mu \quad \text{var}(X) = \sigma^2$$